Edge Theorem in Lean 4 (Lemma 6.1, real case)

TODO: subtitle — polytopic families and exposed edges

TODO: Author Name · Institute

Outline

1. **Background** — Why the Edge Theorem is needed in Control Systems
2. **Preliminaries** — coefficient vectors, polytopes, affine span/direction/dimension, supporting hyperplanes, root spaces
3. **Exposed faces and edges** — boundary geometry and examples
4. **Edge Theorem** — Assumption 6.1, Lemmas 6.1–6.2, theorem, how the lemmas fit
5. **Proof of Lemma 6.1 (real case)** — Steps 1–6 in Lean, recap, complex-case extra
6. **Proof of Lemma 6.2 (real case)** — goal, math proof, helper lemma, Lean walkthroughs, maps, complex-case extra
7. **Edge Theorem assembled** — formal proof, lemmas behind the steps, Lean implementation

What is a control system?

> A **control system** manages, commands, directs, or regulates the behavior of other devices or systems using control loops — from simple applications like a home thermostat to complex industrial processes.

Continuous-time state-space model:

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

Discrete-time state-space model:

$$x[k+1] = A_d x[k] + B_d u[k]$$
$$y[k] = C_d x[k] + D_d u[k]$$

Where: x is the state vector

y the output vector

u the input (control) vector

A, B, C, D the state, input, output, and feedthrough matrices, respectively.

From state-space to transfer function

Starting from $\dot{x}(t) = Ax(t) + Bu(t)$, $y(t) = Cx(t) + Du(t)$, with zero initial conditions ($x(0) = 0$):

Applying Laplace transform:
$sX(s) = AX(s) + BU(s)$, $Y(s) = CX(s) + DU(s)$.
$(sI - A)X(s) = BU(s)$,
so $X(s) = (sI - A)^{-1}BU(s)$.
$\implies Y(s) = C(sI - A)^{-1}BU(s) + DU(s)$.

> $G(s) = \dfrac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$ — the **transfer function**.

Example: mass-spring-damper

$m = 1\,\mathrm{kg}$, $c = 2\,\mathrm{N{\cdot}s/m}$, $k = 3\,\mathrm{N/m}$; input $u(t)$ = force (N), output $y(t)$ = displacement (m).

$$m\ddot{y} + c\dot{y} + ky = u(t) \quad \Rightarrow \quad \ddot{y} + 2\dot{y} + 3y = u(t)$$

States $x_1 = y$ (position), $x_2 = \dot{y}$ (velocity):

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -3x_1 - 2x_2 + u \end{cases} \quad \Rightarrow \quad A = \begin{bmatrix} 0 & 1 \\ -3 & -2 \end{bmatrix},\ B = \begin{bmatrix} 0 \\ 1 \end{bmatrix},\ C = \begin{bmatrix} 1 & 0 \end{bmatrix},\ D = 0$$

$$G(s) = C(sI - A)^{-1}B + D = \begin{bmatrix} 1 & 0 \end{bmatrix} \frac{1}{s^2+2s+3} \begin{bmatrix} s+2 & 1 \\ -3 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{s^2 + 2s + 3}$$

Poles and stability

> **Stability criterion.** A system is **stable** iff all poles of $G(s)$ have strictly negative real parts; it is **unstable** if some pole of $G(s)$ has positive real part.

Example. $G(s) = \dfrac{1}{s^2 + 2s + 3}$: the poles solve $s^2 + 2s + 3 = 0$,

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{-8}}{2} = -1 \pm j\sqrt{2}$$

so with $p_1 = -1 + j\sqrt{2}$ and $p_2 = -1 - j\sqrt{2}$,
$\mathrm{Re}(p_1) = \mathrm{Re}(p_2) = -1 < 0 \Rightarrow$ the system is **stable**.

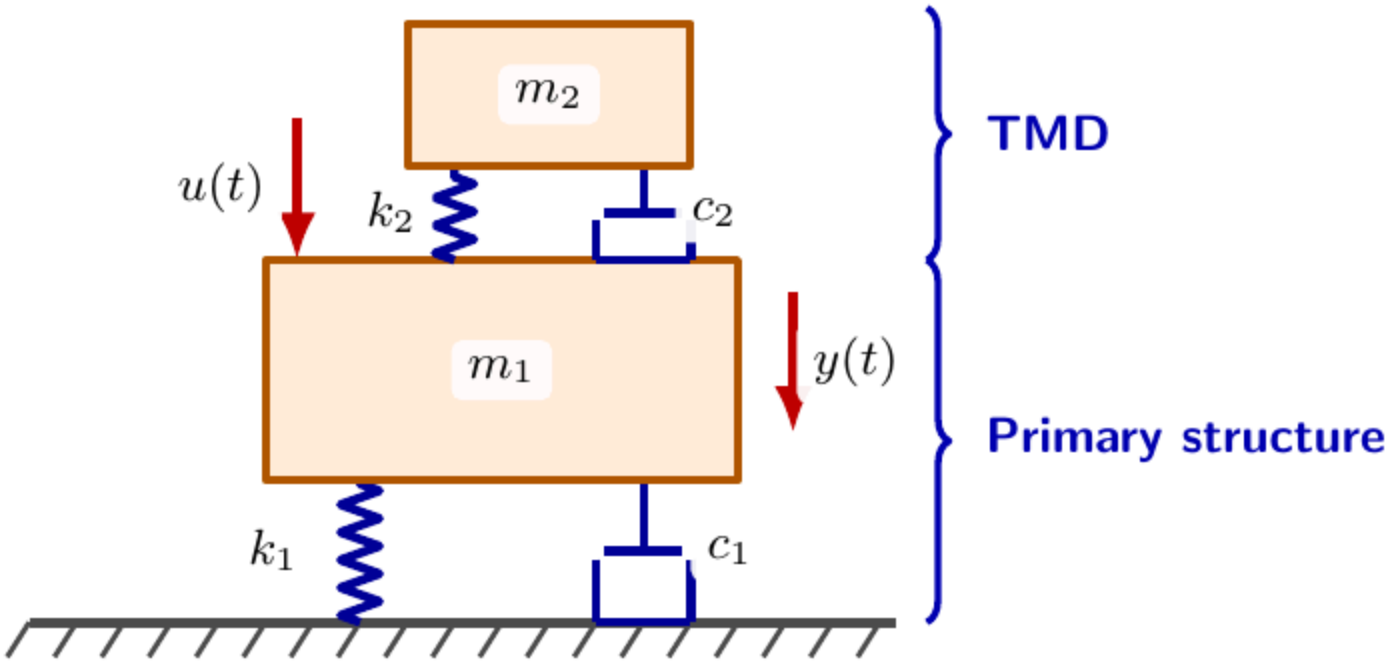
Kharitonov's theorem

Interval family $\mathcal{P} = \{p(s) = a_n s^n + \cdots + a_1 s + a_0 \mid a_i \in [\underline{a}_i, \bar{a}_i],\ \bar{a}_n > 0\}$. Four vertex polynomials:

$$K_1 = \underline{a}_n s^n + \underline{a}_{n-1} s^{n-1} + \bar{a}_{n-2} s^{n-2} + \bar{a}_{n-3} s^{n-3} + \cdots$$
$$K_2 = \bar{a}_n s^n + \bar{a}_{n-1} s^{n-1} + \underline{a}_{n-2} s^{n-2} + \underline{a}_{n-3} s^{n-3} + \cdots$$
$$K_3 = \underline{a}_n s^n + \bar{a}_{n-1} s^{n-1} + \bar{a}_{n-2} s^{n-2} + \underline{a}_{n-3} s^{n-3} + \cdots$$
$$K_4 = \bar{a}_n s^n + \underline{a}_{n-1} s^{n-1} + \underline{a}_{n-2} s^{n-2} + \bar{a}_{n-3} s^{n-3} + \cdots$$

$\mathcal{P}$ Hurwitz stable $\iff$ K_1, K_2, K_3, K_4 each Hurwitz stable.

Example (primary structure + TMD). Primary: $m_1 \in [\underline{m}_1, \bar{m}_1]$, $k_1 \in [\underline{k}_1, \bar{k}_1]$, $c_1 \in [\underline{c}_1, \bar{c}_1]$; tuned mass damper: $m_2 \in [\underline{m}_2, \bar{m}_2]$, $k_2 \in [\underline{k}_2, \bar{k}_2]$, $c_2 \in [\underline{c}_2, \bar{c}_2]$. The equations of motion give $p(s) = a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0 = 0$ with $a_i \in [\underline{a}_i, \bar{a}_i]$ — an interval family, so these four polynomials decide its stability.



Kharitonov vs Edge Theorem (discrete systems)

Why Kharitonov fails (discrete)

- Kharitonov's proof relies on a special even/odd symmetry of the imaginary axis. The unit circle has no such symmetry, so four stable corner polynomials prove nothing — an unstable member can still hide in the family.
- A bilinear map sends the disc to the half-plane, but it mixes all coefficients together

Counterexample: Kharitonov fails (discrete)

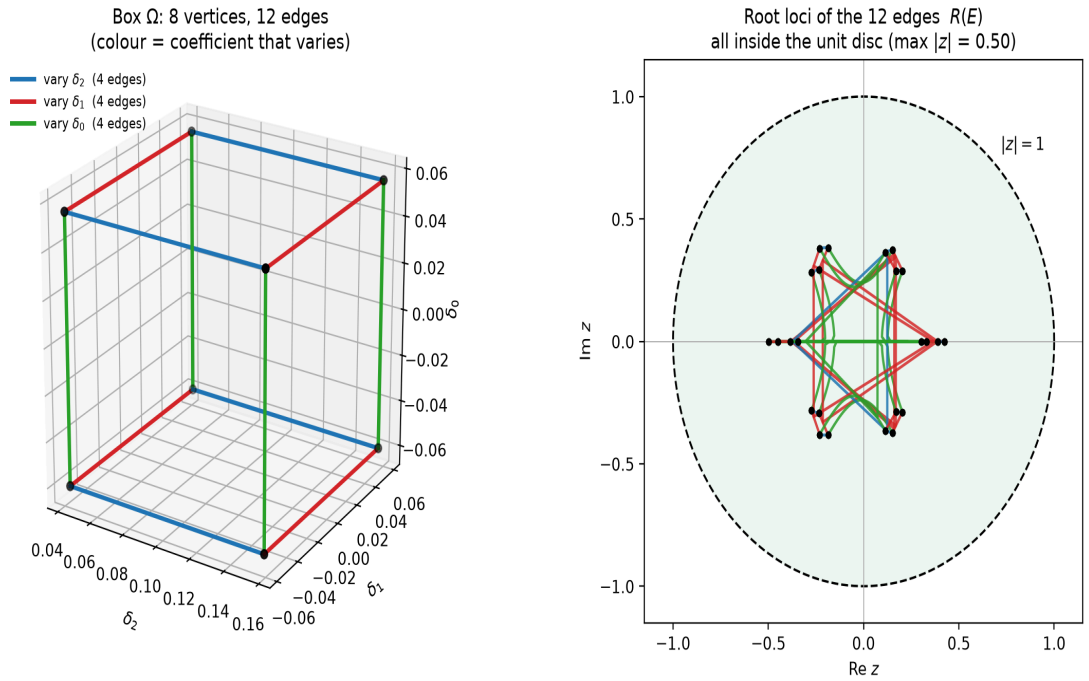
- Interval family $p(z) = a_3 z^3 + a_2 z^2 + a_1 z + a_0$ with $a_0 \in [-0.2, 0.5]$, $a_1 \in [0.3, 0.5]$, $a_2 \in [0.6, 1.3]$, $a_3 \in [0.9, 1.2]$.
- All four Kharitonov polynomials are stable (largest root moduli 0.83, 0.79, 0.98, 0.82 — all < 1).
- But the vertex member $p^\star(z) = 0.9z^3 + 1.3z^2 + 0.3z + 0.5$ has a root of modulus $1.47 > 1$ — unstable. Four stable corners, yet the family is not stable.

Discrete-time robustness needs polytopes, not intervals — and a theorem that works on any stability region.

The Edge Theorem at work (discrete)

Theorem (Edge Theorem; Bartlett–Hollot–Lin 1988). Let Ω be a polytope of fixed-degree polynomials. Then $\partial R(\Omega) \subseteq \bigcup_E R(E)$ over the exposed edges E of Ω. Hence the family is stable (Hurwitz, Schur, ...) iff every exposed edge is — and each edge is a one-parameter family $a(\lambda) = (1-\lambda)a^{(1)} + \lambda a^{(2)}$, $\lambda \in [0,1]$.

Example (Ch. 6, Ex. 6.2). Family $\delta(z;\underline{\delta}) = z^3 + \delta_2 z^2 + \delta_1 z + \delta_0$ with $\underline{\delta} = (\delta_3, \delta_2, \delta_1, \delta_0) \in \Omega := \{1\} \times [0.042, 0.158] \times [-0.058, 0.058] \times [-0.06, 0.056] \subset \mathbb{R}^4$:
a compact 3-dimensional box (8 vertices, 12 edges) with leading coefficient $\equiv 1$.



Ex. 6.2: the box (left) and its 12 edge root loci inside |z| < 1 (right)

Section 1 · Preliminaries

Polytopic family of polynomials

A polytopic family of polynomials can be thought of as the convex hull of a finite number of points (polynomials). Mathematically, this can be represented as the family

$$P(s) = \lambda_1 P_1(s) + \cdots + \lambda_n P_n(s)$$

where $P_i(s)$ are fixed real polynomials and the λ_i are real with $\lambda_i \geq 0$ and $\sum \lambda_i = 1$.

> **Definition (Polytope).**
> Let $v_1, \ldots, v_N \in \mathbb{R}^d$. The *polytope* spanned by $v_1, \ldots, v_N$ is the set of all convex combinations of these points:
>
> $$\Omega \;=\; \operatorname{conv}\{v_1, \ldots, v_N\} \;=\; \left\{ \sum_{i=1}^{N} \lambda_i \, v_i \;\middle|\; \lambda_i \in \mathbb{R},\; \lambda_i \geq 0,\; \sum_{i=1}^{N} \lambda_i = 1 \right\}.$$
>
> Equivalently, Ω is compact and convex.
> (The converse fails: not every compact convex set is a polytope.)
> The *dimension* of Ω is $\dim \Omega := \dim \operatorname{aff}(\Omega)$, the dimension of the smallest affine subspace containing Ω.

Coefficient vectors (`CoeffVec`)

Restatement.

A degree-n polynomial $\delta(s) = \delta_0 + \delta_1 s + \cdots + \delta_n s^n$

is identified with its coefficient vector

$\underline{\delta} = (\delta_0, \delta_1, \ldots, \delta_n) \in \mathbb{R}^{n+1}$.

Lean (`EdgeTheoremDefs.lean`):

EdgeTheoremDefs.lean

```
/-- Coefficient vector `(α₀,…,αₙ) : ℝ^{n+1}` for `δ(s) = α₀+α₁s+…+αₙsⁿ`. -/
abbrev CoeffVec (n : ℕ) := Fin (n + 1) → ℝ
```

EdgeTheoremDefs.lean:95-99

```
/-- Convert a coefficient vector α : Fin(n+1) → ℝ to a polynomial
    δ(s) = α(0) + α(1)·s + ... + α(n)·sⁿ
-/
noncomputable def polyOfVec {n : ℕ} (α : CoeffVec n) : Polynomial ℝ :=
  ∑ j : Fin (n + 1), Polynomial.monomial j.val (α j)
```

Example ($n = 2$).

`CoeffVec 2` = `Fin 3 → ℝ`, i.e. $\mathbb{R}^3$. Take

$\alpha = (\alpha_0, \alpha_1, \alpha_2) = (1,\ -2,\ 3)$, i.e.

$\alpha(0) = 1,\ \alpha(1) = -2,\ \alpha(2) = 3$. Then

$$\texttt{polyOfVec}\ \alpha = 1 + (-2)\, s + 3\, s^2 = 3s^2 - 2s + 1.$$

EdgeTheoremDefs.lean

```
-- α : CoeffVec 2,  α 0 = 1,  α 1 = -2,  α 2 = 3
-- polyOfVec α = 3s² - 2s + 1
```

Polytope: Lean formalization

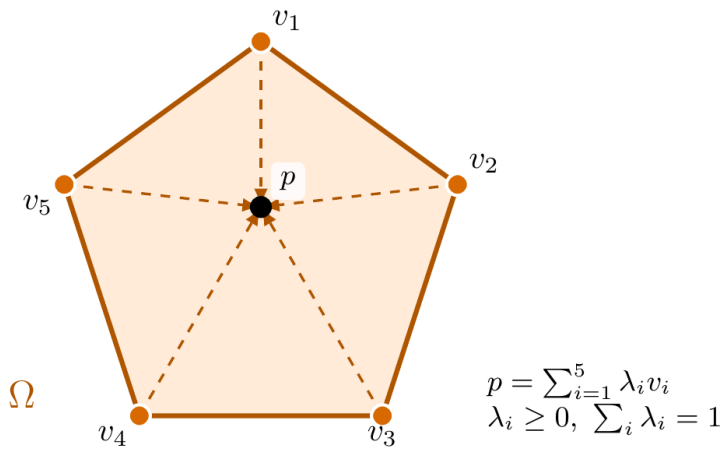
Lean (`EdgeTheoremDefs.lean`):

EdgeTheoremDefs.lean

```
/--
A polytope Ω in coefficient space ℝ^{n+1}.
It is defined as the convex hull of a finite set of vertices V,
and is assumed to have nonempty topological interior.
-/
structure Polytope (n : ℕ) where
  vertices : Finset (CoeffVec n)
  nonempty  : vertices.Nonempty
  interior_nonempty : (interior (convexHull ℝ (vertices : Set (CoeffVec n)))).Nonempty

/-- The actual set Ω ⊆ ℝ^{n+1} as the convex hull of the vertices. -/
def Polytope.Ω (P : Polytope n) : Set (CoeffVec n) :=
  convexHull ℝ (P.vertices : Set (CoeffVec n))
```

> **Assumption.** The Lean `Polytope` structure additionally requires nonempty interior (`interior_nonempty`). Everything in this deck assumes it.



Affine span, direction, dimension — definition

> **Definition (Affine subspace).** $A \subseteq \mathbb{R}^d$ is affine iff $\forall x, y \in A,\ \forall t \in \mathbb{R},\ (1-t)x + ty \in A$: closed under lines. Equivalently $A = p + V$ for $p \in A$ and a linear subspace $V = \mathrm{dir}(A)$.

> **Definition (Affine span).** For $S \subseteq \mathbb{R}^d$, $\mathrm{aff}(S) = \bigcap \{A \text{ affine} \mid S \subseteq A\}$, the smallest affine subspace containing S.

> **Definition (Direction).** For an affine subspace A, $\mathrm{dir}(A) = \{v \mid \forall p \in A,\ p + v \in A\}$, the underlying linear subspace of translations preserving A.

> **Definition (Dimension).** For a linear subspace $U \subseteq \mathbb{R}^d$, $\dim U$ is the number of vectors in any basis of U, i.e. the maximal size of a linearly independent subset of U. For an affine subspace A, $\dim A := \dim \mathrm{dir}(A)$. Hence $\dim \mathrm{aff}(F) = 0$ means a point, $= 1$ a line, $= 2$ a plane.

Lean (`EdgeTheoremDefs.lean:341-354`): continued on the next slide.

Affine span, direction, dimension — Lean

EdgeTheoremDefs.lean:338-351

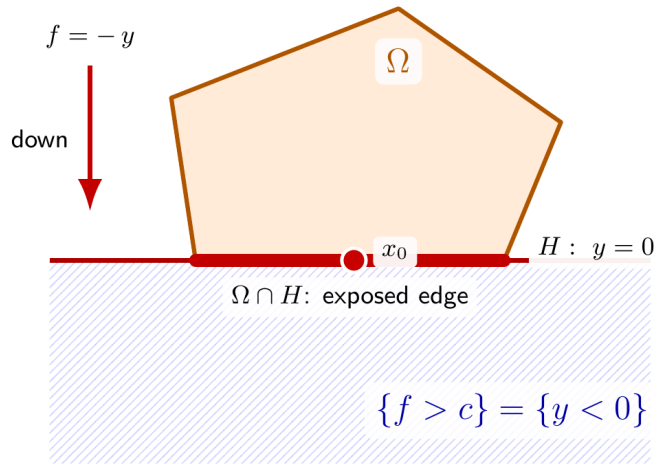
```
abbrev meetDir (n : ℕ) (U : Submodule ℝ (CoeffVec n))
    (affΩ : AffineSubspace ℝ (CoeffVec n)) : Submodule ℝ (CoeffVec n) :=
  (affineSpan ℝ ((U : Set (CoeffVec n)) ∩ (affΩ : Set (CoeffVec n)))).direction


noncomputable abbrev dim {M : Type*} [AddCommGroup M] [Module ℝ M]
    (U : Submodule ℝ M) : ℕ :=
  Module.finrank ℝ U
```

Supporting hyperplane

Definition (Supporting hyperplane).
Let $\Omega \subseteq \mathbb{R}^d$ be convex and
$f : \mathbb{R}^d \to \mathbb{R}$ a nonzero linear functional. The affine set
$H = \{\, x \in \mathbb{R}^d : f(x) = c \,\}$ is a *supporting hyperplane* of Ω if
(i) $f(x) \leq c$ for all $x \in \Omega$
(ii) $f(x_0) = c$ for some $x_0 \in \Omega$.
The *exposed set* is $\Omega \cap H$.

Example (Concrete supporting hyperplane).
Take $f(x, y) = -y$ and $c = 0$.
$H = \{\, (x, y) : y = 0 \,\}$
is the x-axis. The polytope Ω lies above it, so
$f(x, y) = -y \leq 0 = c$ on Ω, and the point
x_0 belongs to both Ω and H.
Hence H supports Ω, and the exposed set $\Omega \cap H$ is the bottom edge of the polytope.



Example $f = -y$, $c = 0$: Ω above $H = \{y = 0\}$; $\Omega \cap H$ is the bottom edge

Supporting hyperplane: Lean

Lean (`EdgeTheoremDefs.lean`):

EdgeTheoremDefs.lean

```
structure SupportingHyperplane (P : Polytope n) where
  f : CoeffVec n →ₗ[ℝ] ℝ
  c : ℝ
  nonzero : f ≠ 0
  upper_bound : ∀ x ∈ P.Ω, f x ≤ c
  touches : ∃ x ∈ P.Ω, f x = c
```

Root spaces

Root space — definition

Notation.
For $\delta = (\delta_0, \ldots, \delta_n) \in \mathbb{R}^{n+1}$,
$\mathrm{polyOfVec}(\delta)(s) = \delta_0 + \delta_1 s + \cdots + \delta_n s^n$.

> **Definition (Root space).**
> Let $W \subseteq \mathbb{R}^{n+1}$. Then
>
> $$R(W) = \{\, s \in \mathbb{C} \mid \exists\, \delta \in W,\ \delta(s) = 0 \,\},$$
>
> and for a polytope P, $R(P) := R(P.\Omega)$.

Lean (`EdgeTheoremDefs.lean`):

EdgeTheoremDefs.lean

```
def RootSpaceSet {n : ℕ}
  (W : Set (CoeffVec n)) : Set ℂ :=
  { s | ∃ δ ∈ W,
      ((polyOfVec δ).map (algebraMap ℝ ℂ)).IsRoot s }

/--
The root space of a polytope `P`: all complex numbers `s` such that
some coefficient vector in `P.Ω` yields a polynomial vanishing at `s`.
-/
def RootSpace (P : Polytope n) : Set ℂ :=
  RootSpaceSet P.Ω
```

Exposed faces and edges

Boundary and exposed faces — definition

Definition (Closure and boundary).
$\overline{S} = \bigcap\{\, F \supseteq S : F \text{ closed} \,\}$ (smallest closed set containing S);
$\partial S = \overline{S} \setminus \operatorname{int}(S)$.

Definition (Intrinsic interior / relative boundary).
For $S \subseteq \mathbb{R}^d$, the *intrinsic interior*
$\operatorname{relint}(S)$ is the interior of S relative to
its affine span: $x \in \operatorname{relint}(S)$ iff
$x \in S$ and $\exists\, \varepsilon > 0$ with
$B(x, \varepsilon) \cap \operatorname{aff}(S) \subseteq S$.
The *relative boundary* is $S \setminus \operatorname{relint}(S)$.

Definition (Exposed face / edge).
For H supporting Ω, $F = \Omega \cap H$ is
exposed; $\dim \operatorname{aff}(F) = 1$ gives an
exposed edge E, $= 2$ an *exposed face*.

Lean (`EdgeTheoremDefs.lean:140-151`)

Boundary and exposed faces — Lean

EdgeTheoremDefs.lean:140-151

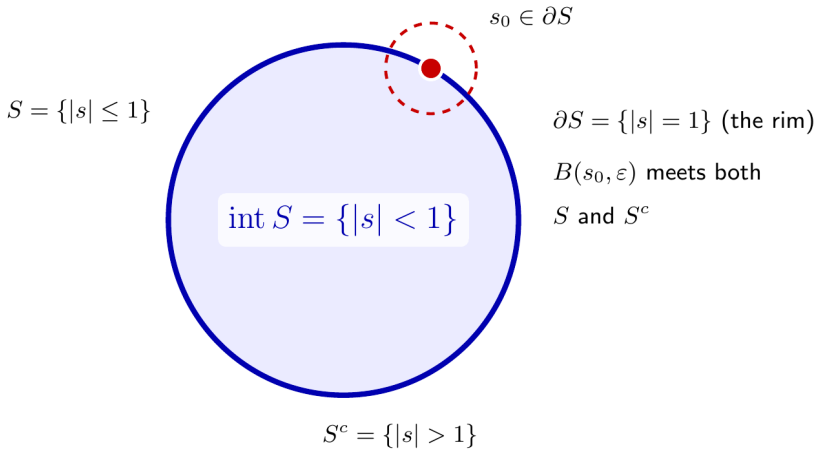
```
/-- The exposed face of `P` for `hp`: `F = Ω ∩ {x | hp.f x = hp.c}`. -/
def ExposedFace {n : ℕ} {P : Polytope n} (hp : SupportingHyperplane P) :
    Set (CoeffVec n) :=
  { x | x ∈ P.Ω ∧ hp.f x = hp.c }

open Affine

/-- `E` is an exposed edge of `P` if it is an exposed face of affine dimension 1. -/
def IsExposedEdge {n : ℕ} (P : Polytope n) (E : Set (CoeffVec n)) : Prop :=
  ∃ hp : SupportingHyperplane P,
    E = ExposedFace hp ∧
    Module.finrank ℝ (affineSpan ℝ (ExposedFace hp)).direction = 1
```

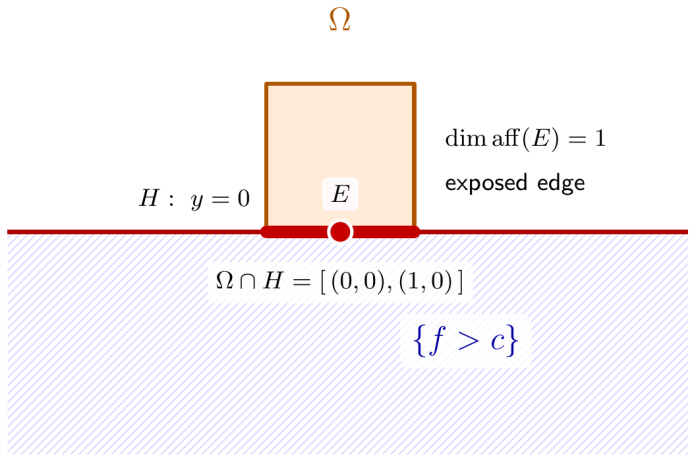
Boundary and exposed edge — example

Example (Boundary).
$S = \{\, s : |s| \le 1 \,\}$ is closed, so $\overline{S} = S$, and $\operatorname{int}(S) = \{\, s : |s| < 1 \,\}$. Thus $\partial S = \{\, s : |s| = 1 \,\}$.



Boundary: $\operatorname{int}(S) = \{|s| < 1\}$, $\partial S = \{|s| = 1\}$ — a small ball about s_0 meets both S and S^c

Example (Exposed edge).
Let $\Omega = \operatorname{conv}\{(0,0), (1,0), (1,1), (0,1)\}$ (unit square) and $H = \{(x, y) : -y = 0\}$. Then $\Omega \cap H = [\,(0,0),\ (1,0)\,]$ with $\dim \operatorname{aff}(\Omega \cap H) = 1$: an exposed edge.



Exposed edge: $H = \{y = 0\}$ cuts the unit square in the segment $[\,(0,0), (1,0)\,]$, of affine dimension 1

Edge Theorem

Assumption 6.1

> **Assumption 6.1.**
> The sign of the leading coefficient δ_n is constant on Ω; e.g. $\delta_n > \Delta > 0$ for all $\underline{\delta} \in \Omega$.

Every lemma and theorem below assumes it.

Lemmas 6.1–6.2 — statements

Lemma 6.1.
Let $\Omega \subset \mathbb{R}^{n+1}$ be a polytope satisfying Assumption 6.1. Then

1. If $s_r \in \mathbb{R}$ and $s_r \in R(\Omega)$, then $\exists$ an exposed edge E of Ω with $s_r \in R(E)$;
2. If $s_c \in \mathbb{C}$ and $s_c \in R(\Omega)$, then $\exists$ an exposed face F of Ω with $s_c \in R(F)$.

Lemma 6.2.
For any exposed face F of Ω,

$$\partial R(F) \subseteq R(\partial F),$$

i.e. the boundary of the root space of F is contained in the root space of the relative boundary of F.

Edge Theorem — statement

> **Theorem (Edge Theorem).**
> Let $\Omega \subset \mathbb{R}^{n+1}$ be a polytope of polynomials satisfying Assumption 6.1. Then the boundary of $R(\Omega)$ is contained in the root space of the exposed edges of Ω:
>
> $$\partial R(\Omega) \subseteq \bigcup_{E \text{ exposed edge}} R(E).$$

Edge Theorem — proof outline

1. **Lemma 6.1.** Every $s \in R(\Omega)$ lies in $R(F)$ for some exposed face F of Ω; if $s = s_r \in \mathbb{R}$ then $F = E$ can be taken to be an exposed edge.
2. **Lemma 6.2.** For F exposed, $\partial R(F) \subseteq R(\operatorname{rel}\partial F)$: a boundary root of a face is already a root on its relative boundary.
3. **Descent.** Let $s \in \partial R(\Omega)$. By (1), $s \in R(F_0)$ for F_0 exposed; while $s \in \partial R(F_k)$, (2) moves the root to $R(F_{k+1})$ for $F_{k+1} \subsetneq F_k$ exposed, so $\dim F_{k+1} < \dim F_k$. The dimension drops strictly at each step, hence the process terminates at an exposed edge E with $s \in R(E)$.

Together: $\partial R(\Omega) \subseteq \bigcup_E R(E)$ over exposed edges E.

Section 4 · Proof of Lemma 6.1, real case

Lemma 6.1 (real case) — proof overview

> **Lemma 6.1.** : Let $\Omega \subset \mathbb{R}^{n+1}$ be a polytope satisfying Assumption 6.1. Then
>
> 1. If $s_r \in \mathbb{R}$ and $s_r \in R(\Omega)$, then $\exists$ an exposed edge E of Ω with $s_r \in R(E)$;
> 2. If $s_c \in \mathbb{C}$ and $s_c \in R(\Omega)$, then $\exists$ an exposed face F of Ω with $s_c \in R(F)$.

1. **Unpack.** $s \in R(\Omega)$ with $\mathrm{Im}(s) = 0$ gives $\delta \in \Omega \cap P_{s_r}(r)$ with $s = \bar{r}$.
2. **Count.** $\dim U = n$ and $\dim \mathrm{dir} \geq 2$ give $\dim(U \cap \mathrm{dir}) \geq 1$.
3. **Escape.** Follow the nonzero direction to $\delta_{\mathrm{bound}} \in P_{s_r} \cap \partial\Omega$.
4. **Expose.** Hahn–Banach at δ_{bound} gives exposed $F \ni \delta_{\mathrm{bound}}$ with $\bar{r} \in R(F)$.
5. **Descend.** $F \to G \to \cdots \to E$ with $\bar{r} \in R(E)$, $\dim E = 1$ (termination by dimension).
6. **Degenerate.** $m = 0$ is impossible; $m = 1$ makes Ω itself the edge.

Step 1: Unpack the root — Lean + map

Claim. Let $s \in R(\Omega)$ with $\mathrm{Im}(s) = 0$. Then $s = \bar{r} \in R$ and there is $\delta \in P_{s_r}(r)$ with $\tilde{P}_\delta(s) = 0$.

Proof. $s \in R(\Omega)$ gives $\delta \in \Omega$ with $\tilde{P}_\delta(s) = 0$.

$\mathrm{Im}(s) = 0$ gives $s = \bar{r}$ ($r = \mathrm{Re}(s)$) by `Complex.ext`.

The root at $\bar{r}$ reads $P_\delta(r) = 0$, i.e. $\delta \in \ker(\mathrm{ev}_r) = P_{s_r}(r)$.

Lemma61.lean:142-153

```
theorem lemma61_real (hn : n ≥ 1) (P : Polytope n) (s : ℂ) (hs : s ∈ RootSpace P) :
    s.im = 0 → ∃ E, IsExposedEdge P E ∧ s ∈ RootSpaceSet E := by
  intro hreal
  unfold RootSpace RootSpaceSet at hs                          1.1
  obtain ⟨δ, hδ_in_Ω, hδ_root⟩ := hs
  have hs_real : s = ↑s.re := by                               1.2
    apply Complex.ext
    · simp
    · simp [hreal]
  have hδ_in_Psr : δ ∈ (P_sr n s.re : Set (CoeffVec n)) := by   1.3
    rw [hs_real] at hδ_root
    exact mem_P_sr_of_isRoot s.re δ hδ_root
```

Map

Step	Content
1.1	$s \in R(\Omega)$ **gives** $\delta \in \Omega$ **with** $\tilde{P}_\delta(s) = 0$
1.2	$\mathrm{Im}(s) = 0$ **gives** $s = \bar{r}$ **with** $r = \mathrm{Re}(s)$
1.3	**Root at** $\bar{r}$ **gives** $P_\delta(r) = 0$**, i.e.** $\delta \in P_{s_r}(r)$

Step 2: Dimension count — Lean + map

Claim. Let U be linear with $\dim U = n$, $\mathrm{aff}\Omega$ affine with $\dim(\mathrm{dir}(\mathrm{aff}\Omega)) \geq 2$, and $\delta \in U \cap \mathrm{aff}\Omega$. Then $\dim(\mathrm{meetDir}) \geq 1$.
Proof. $\mathrm{meetDir} = \mathrm{dir}(\mathrm{aff}(U \cap \mathrm{aff}\Omega))$ by definition.
2.2 Put $\mathrm{Aint} = U \sqcap \mathrm{aff}\Omega$.
2.3 $\mathrm{dir}(\mathrm{Aint}) = U \cap \mathrm{dir}(\mathrm{aff}\Omega)$, Since U, viewed as an affine subspace, has direction U itself
2.4 $\mathrm{aff}(U \cap \mathrm{aff}\Omega) = \mathrm{Aint}$.
2.5 Hence $\dim(\mathrm{meetDir}) = \dim(U \cap \mathrm{dir}(\mathrm{aff}\Omega))$.
2.6 $\dim(U \cap V) = \dim U + \dim V - \dim(U + V) \geq n + 2 - (n + 1) = 1$, where V is $\mathrm{dir}(\mathrm{aff}\Omega)$

PreliminaryLemmas.lean

```
lemma intersection_affine_dim_ge_one {n : ℕ} (U : Submodule ℝ (CoeffVec n))
    (affΩ : AffineSubspace ℝ (CoeffVec n))
    (δ : CoeffVec n) (hδU : δ ∈ U) (hδΩ : δ ∈ affΩ)
    (hU_dim : dim U = n) (haff_dim : dim affΩ.direction ≥ 2) :
    dim (meetDir n U affΩ) ≥ 1 := by
  unfold meetDir                                                    2.1
  let Aint : AffineSubspace ℝ (CoeffVec n) := U.toAffineSubspace ⊓ affΩ   2.2
  have hA_dir : Aint.direction = U ⊓ affΩ.direction :=               2.3
    intersection_direction_eq U affΩ δ hδU hδΩ
  have hA_eq : affineSpan ℝ ((U : Set (CoeffVec n)) ∩ (affΩ : Set (CoeffVec n))) = Aint := by   2.4
    rw [affineSpan_inter U affΩ]
  rw [hA_eq, hA_dir]                                                2.5
  exact finrank_inf_ge_one U affΩ.direction hU_dim haff_dim          2.6
```

Map

Step	Content
2.1	$\mathrm{meetDir} = \mathrm{dir}(\mathrm{aff}(U \cap \mathrm{aff}\Omega))$ **by definition**
2.2	$\mathrm{Aint} = U \sqcap \mathrm{aff}\Omega$ **(infimum of affine subspaces)**
2.3	$\mathrm{dir}(\mathrm{Aint}) = U \cap \mathrm{dir}(\mathrm{aff}\Omega)$
2.4	$\mathrm{aff}(U \cap \mathrm{aff}\Omega) = \mathrm{Aint}$ **(a subspace is affine)**
2.5	**Hence** $\dim(\mathrm{meetDir}) = \dim(U \cap \mathrm{dir}(\mathrm{aff}\Omega))$
2.6	$\dim(U \cap V) = \dim U + \dim V - \dim(U + V) \geq n + 2 - (n + 1) = 1$,

Step 3: Escape — Lean + map

Claim. Let Ω be a polytope, U a linear subspace, $\delta \in \Omega \cap U$, with $\dim(\mathrm{meetDir}) \geq 1$. Then $\exists\, \delta_{\mathrm{bound}} \in U \cap \partial\Omega$.

Proof. $\dim \geq 1$ gives $v \neq 0$ in $U \sqcap \mathrm{dir}(\mathrm{aff}\Omega)$.

$\delta + tv \in U$ for all t because U is a subspace containing both δ and v; $\delta + tv \in \mathrm{aff}\Omega$ because an affine space contains the whole line through any of its points in any of its directions (line_in_intersection).

Ω is compact, hence bounded: all pairs lie at distance $\leq C$. With $|v| > 0$ (from ①), $t_{\mathrm{out}} := (|C| + 1)/|v| > 0$, and $\mathrm{dist}(\delta + t_{\mathrm{out}} v,, \delta) = |C| + 1 > C$, so the point lies outside Ω (ray_escapes_of_bounded).

The segment $[\delta, \delta + t_{\mathrm{out}} v]$ meets $\partial\Omega$.

$\delta_{\mathrm{bound}} = \delta + (c \cdot t_{\mathrm{out}})v \in U \cap \partial\Omega$.

```
lemma exists_boundary_point_in_submodule {n : ℕ} (P : Polytope n)
    (U : Submodule ℝ (CoeffVec n)) (δ : CoeffVec n)
    (hδ_in_Ω : δ ∈ P.Ω) (hδ_in_U : δ ∈ (U : Set (CoeffVec n)))
    (affΩ : AffineSubspace ℝ (CoeffVec n)) (hδ_aff : δ ∈ affΩ)
    (hA_dim : dim (meetDir n U affΩ) ≥ 1) :
    ∃ δ_bound, δ_bound ∈ (U : Set (CoeffVec n)) ∩ frontier P.Ω := by
  have h_dim_pos : 0 < dim (meetDir n U affΩ) := by omega
  haveI : Nontrivial ↥(U ⊓ affΩ.direction) :=
    intersection_nontrivial U affΩ δ hδ_in_U hδ_aff h_dim_pos
  obtain ⟨v_sub, hv_sub_nonzero⟩ := exists_ne (0 : ↥(U ⊓ affΩ.direction))                    ①

  let v : CoeffVec n := v_sub.val
  have h_line_in_intersection : ∀ (t : ℝ), δ + t • v ∈ (U : Set (CoeffVec n)) ∩
      (affΩ : Set (CoeffVec n)) :=
    line_in_intersection U affΩ δ hδ_in_U hδ_aff v_sub                                      ②

  have hv_nonzero : v ≠ 0 := by
    intro h; apply hv_sub_nonzero; exact Submodule.coe_eq_zero.mp h
  have h_escapes : ∃ (t : ℝ), 0 < t ∧ δ + t • v ∉ P.Ω :=
    ray_escapes_of_bounded P.isBounded δ v hδ_in_Ω hv_nonzero
  obtain ⟨t_out, ht_out_pos, ht_out⟩ := h_escapes                                           ③

  by_cases hδ_front : δ ∈ frontier P.Ω
  · exact ⟨δ, hδ_in_U, hδ_front⟩
  · obtain ⟨δ_bound, h_seg, h_front⟩ :=
      segment_boundary_intersection P δ hδ_in_Ω hδ_front v hv_nonzero t_out ht_out           ④

    rw [segment_eq_image] at h_seg
    obtain ⟨c, hc_in_Icc, hc_eq⟩ := h_seg
    have h_rewrite : (1 - c) • δ + c • (δ + t_out • v) = δ + (c * t_out) • v :=
      segment_point_rewrite δ v c t_out
    have h_mem := h_line_in_intersection (c * t_out)
    rcases h_mem with ⟨hmem_U, hmem_aff⟩
    have h_eq : δ_bound = δ + (c * t_out) • v := by
      calc δ_bound = (1 - c) • δ + c • (δ + t_out • v) := by rw [←hc_eq]
        _ = δ + (c * t_out) • v := by rw [h_rewrite]
    exact ⟨δ_bound, h_eq ▸ hmem_U, h_front⟩                                                 ⑤
```

Map

Step	Content
①	$\dim \geq 1$ **makes** $U \sqcap \mathrm{dir}(\mathrm{aff}\Omega)$ **nontrivial, giving** $v \neq 0$
②	$\delta + tv$ **stays in** $U \cap \mathrm{aff}\Omega$: U **is a subspace,** $\mathrm{aff}\Omega$ **is affine**
③	**Bounded** Ω **gives** $t_{\mathrm{out}} = (\lvert C\rvert + 1)/\lvert v\rvert > 0$ **with** $\delta + t_{\mathrm{out}} v \notin \Omega$ **(distance** $\lvert C\rvert + 1 > C$**)**
④	$[\delta, \delta + t_{\mathrm{out}} v]$ **meets** $\partial\Omega$ **(or** $\delta \in \partial\Omega$**); else two disjoint opens disconnect it**
⑤	$\delta_{\mathrm{bound}} = \delta + (c \cdot t_{\mathrm{out}})v$**: on the line from ②, hence in** U **— and in** $\partial\Omega$ **by ④**

Step 3: Escape — `exists_boundary_point_in_submodule` (Lean + map)

Goal. Let $\Omega \subset \mathbb{R}^{n+1}$ be a polytope, U a linear subspace, $\delta \in \Omega \cap U$, with $\dim(\mathrm{meetDir}\, n\, U\, \mathrm{aff}\Omega) \geq 1$. Then $\exists\, \delta_{\mathrm{bound}} \in U \cap \partial\Omega$.

Proof. 3.1 $\dim \geq 1$ makes $U \sqcap \mathrm{dir}(\mathrm{aff}\Omega)$ nontrivial, giving $v_{\mathrm{sub}} \neq 0$.

3.2 With $v := v_{\mathrm{sub}}.\mathrm{val}$, the line $\delta + t \cdot v$ lies in $U \cap \mathrm{aff}\Omega$ for all t.

3.3 Boundedness of Ω gives $t_{\mathrm{out}} > 0$ with $\delta + t_{\mathrm{out}} v \notin \Omega$ (and $v \neq 0$ is inherited).

3.4 If $\delta \in \partial\Omega$ take $\delta_{\mathrm{bound}} = \delta$; else the segment $[\delta,\ \delta + t_{\mathrm{out}} v]$ meets $\partial\Omega$.

3.5 Rewriting gives $\delta_{\mathrm{bound}} = \delta + (c \cdot t_{\mathrm{out}})v \in U \cap \partial\Omega$.

PreliminaryLemmas.lean:336-368 — full proof, 3.1–3.5 highlighted

```
lemma exists_boundary_point_in_submodule {n : ℕ} (P : Polytope n)
    (U : Submodule ℝ (CoeffVec n)) (δ : CoeffVec n)
    (hδ_in_Ω : δ ∈ P.Ω) (hδ_in_U : δ ∈ (U : Set (CoeffVec n)))
    (affΩ : AffineSubspace ℝ (CoeffVec n)) (hδ_aff : δ ∈ affΩ)
    (hA_dim : dim (meetDir n U affΩ) ≥ 1) :
    ∃ δ_bound, δ_bound ∈ (U : Set (CoeffVec n)) ∩ frontier P.Ω := by
  have h_dim_pos : 0 < dim (meetDir n U affΩ) := by omega                    -- 3.1
  haveI : Nontrivial ↥(U ⊓ affΩ.direction) :=
    intersection_nontrivial U affΩ δ hδ_in_U hδ_aff h_dim_pos
  obtain ⟨v_sub, hv_sub_nonzero⟩ := exists_ne (0 : ↥(U ⊓ affΩ.direction))
  let v : CoeffVec n := v_sub.val                                            -- 3.2
  have h_line_in_intersection : ∀ (t : ℝ), δ + t • v ∈ (U : Set (CoeffVec n)) ∩
      (affΩ : Set (CoeffVec n)) :=
    line_in_intersection U affΩ δ hδ_in_U hδ_aff v_sub
  have hv_nonzero : v ≠ 0 := by                                              -- 3.3
    intro h; apply hv_sub_nonzero; exact Submodule.coe_eq_zero.mp h
  have h_escapes : ∃ (t : ℝ), 0 < t ∧ δ + t • v ∉ P.Ω :=
    ray_escapes_of_bounded P.isBounded δ v hδ_in_Ω hv_nonzero
  obtain ⟨t_out, ht_out_pos, ht_out⟩ := h_escapes
  by_cases hδ_front : δ ∈ frontier P.Ω                                        -- 3.4
  · exact ⟨δ, hδ_in_U, hδ_front⟩
  · obtain ⟨δ_bound, h_seg, h_front⟩ :=
      segment_boundary_intersection P δ hδ_in_Ω hδ_front v hv_nonzero t_out ht_out
    rw [segment_eq_image] at h_seg                                           -- 3.5
    obtain ⟨c, hc_in_Icc, hc_eq⟩ := h_seg
    have h_rewrite : (1 - c) • δ + c • (δ + t_out • v) = δ + (c * t_out) • v :=
      segment_point_rewrite δ v c t_out
    have h_mem := h_line_in_intersection (c * t_out)
    rcases h_mem with ⟨hmem_U, hmem_aff⟩
    have h_eq : δ_bound = δ + (c * t_out) • v := by
      calc δ_bound = (1 - c) • δ + c • (δ + t_out • v) := by rw [←hc_eq]
        _ = δ + (c * t_out) • v := by rw [h_rewrite]
    exact ⟨δ_bound, h_eq ▸ hmem_U, h_front⟩
```

Map

Step	Content
3.1	`dim(meetDir) ≥ 1 → U ⊓ dir` **nontrivial → nonzero** `v`
3.2	**Line** `δ + t • v ⊆ U ∩ affΩ`
3.3	`∃ t_out > 0, δ + t_out • v ∉ Ω` (`t_out = (|C|+1)/‖v‖`)
3.4	**Segment** `[δ, δ + t_out • v]` **crosses** `∂Ω` **(connectedness)**
3.5	**The witness stays in the slice:** δ_{bound} **on the segment gives** $c \in [0,1]$ **with** $\delta_{\mathrm{bound}} = (1-c)\delta + c(\delta + t_{\mathrm{out}} v) = \delta + (c \cdot t_{\mathrm{out}})v$ **(via** `segment_eq_image` , `segment_point_rewrite` **), a point of the line from 3.2, hence in** U **— and in** $\partial\Omega$ **by 3.4**

Step 3: Escape — `ray_escapes_of_bounded` (Lean + map)

BasicLemmas.lean:73-91 — full proof, A.1.1–A.1.4 highlighted

```
lemma ray_escapes_of_bounded {n : ℕ} {S : Set (CoeffVec n)}
    (hB : Bornology.IsBounded S) (δ v : CoeffVec n)
    (hδ : δ ∈ S) (hv : v ≠ 0) : ∃ t : ℝ, 0 < t ∧ δ + t • v ∉ S := by
  rcases Metric.isBounded_iff.mp hB with ⟨C, hC⟩                    A.1.1
  have hv_norm_pos : 0 < ‖v‖ := norm_pos_iff.mpr hv
  refine (((|C| + 1) / ‖v‖, div_pos (by positivity) hv_norm_pos, ?_)   A.1.2
  intro h_contra
  have ht_nonneg : (0 : ℝ) ≤ (|C| + 1) / ‖v‖ := (div_pos (by positivity) hv_norm_pos).le
  have h_dist : dist (δ + (((|C| + 1) / ‖v‖) • v) δ = (((|C| + 1) / ‖v‖) * ‖v‖ := by
    rw [dist_eq_norm]
    have h_sub : δ + (((|C| + 1) / ‖v‖) • v - δ = (((|C| + 1) / ‖v‖) • v := by abel
    rw [h_sub, norm_smul, Real.norm_eq_abs _,
      abs_of_nonneg ht_nonneg]                                      A.1.3
  have h_le : dist (δ + (((|C| + 1) / ‖v‖) • v) δ ≤ C := hC h_contra hδ
  have h_t_mul : (((|C| + 1) / ‖v‖) * ‖v‖ = |C| + 1 :=
    div_mul_cancel₀ _ (ne_of_gt hv_norm_pos)
  rw [h_dist, h_t_mul] at h_le
  have : C ≤ |C| := le_abs_self C                                   A.1.4
  linarith
```

Map

Step	Content
A.1.1	**Boundedness gives radius C:** $\mathrm{dist} \le C$ **on** S
A.1.2	$t_{out} = (\lvert C\rvert + 1)/\lvert v\rvert > 0$ **witnesses the existential**
A.1.3	$\mathrm{dist} = t\lvert v\rvert = \lvert C\rvert + 1$ (`dist_eq_norm`, `abel`, `norm_smul`, `div_mul_cancel₀`)
A.1.4	$\lvert C\rvert + 1 \le C \le \lvert C\rvert$**, impossible**

Step 3: Escape — `segment_boundary_intersection` (Lean + map)

Claim. Let $\delta \in P.\Omega \setminus \text{frontier } P.\Omega$, $v \neq 0$, and t_{out} with $\delta + t_{\text{out}} \bullet v \notin P.\Omega$. Then $\exists\, \delta_{\text{bound}} \in \text{segment}[\delta\,, (\delta + t_{\text{out}} \bullet v)\,]$ with $\delta_{\text{bound}} \in \text{frontier } P.\Omega$.

Proof. A.2.1 Assume Ω is closed and prove by contradiction (h_no_front : ∀ δ_bound ∈ segment ℝ δ (δ + t_out • v), δ_bound ∉ frontier P.Ω , and goal becomes false)

A.2.2 The segment is then covered by the union of the two opens $\text{int}\,\Omega$ and $\text{int}(\Omega^c)$. Points in Ω off the frontier are interior since Ω is closed ($\partial\Omega = \Omega \setminus \text{int}\,\Omega$), and points outside Ω lie in the open complement, hence in its interior.

A.2.3 Both opens meet the segment: δ lies in $\text{int}\,\Omega$ (in Ω yet off the frontier), and $\delta + t_{\text{out}} v$ lies in $\text{int}(\Omega^c)$ (outside the closed Ω); moreover the two opens are disjoint, $\text{int}\,\Omega \cap \text{int}(\Omega^c) = \emptyset$.

A.2.4 Connectedness of the segment forces a point in both — inside $\emptyset$, contradiction.

`PreliminaryLemmas.lean:273-294 — A.2.1–A.2.4 highlighted`

```
lemma segment_boundary_intersection {n : ℕ} (P : Polytope n) (δ : CoeffVec n)
    (hδ_in_Ω : δ ∈ P.Ω) (hδ_not_front : δ ∉ frontier P.Ω)
    (v : CoeffVec n) (_hv_nonzero : v ≠ 0) (t_out : ℝ) (ht_out : δ + t_out • v ∉ P.Ω) :
    ∃ δ_bound ∈ segment ℝ δ (δ + t_out • v), δ_bound ∈ frontier P.Ω := by
  have h_conn : IsConnected (segment ℝ δ (δ + t_out • v)) :=                    A.2.4
    segment_is_connected δ v t_out
  have h_closed : IsClosed P.Ω := P.isCompact.isClosed                          A.2.1
  by_contra h_no_front
  push_neg at h_no_front
  have h_cover : segment ℝ δ (δ + t_out • v) ⊆ (interior P.Ω) ∪ interior (P.Ωᶜ) := A.2.2
    segment_cover_by_interior_and_complement P δ v t_out h_closed h_no_front
  have h_in_u : (segment ℝ δ (δ + t_out • v) ∩ interior P.Ω).Nonempty :=       A.2.3
    segment_intersects_interior P δ v t_out hδ_in_Ω hδ_not_front h_closed
  have h_in_v : (segment ℝ δ (δ + t_out • v) ∩ interior (P.Ωᶜ)).Nonempty :=
    segment_intersects_complement_interior P δ v t_out ht_out h_closed
  have huv_empty : interior P.Ω ∩ interior (P.Ωᶜ) = ∅ :=
    interior_and_complement_interior_disjoint P
  have h_pre := h_conn.2 (interior P.Ω) (interior (P.Ωᶜ)) isOpen_interior isOpen_interior   A.2.4
  have h_inter_nonempty := h_pre h_cover h_in_u h_in_v
  obtain ⟨x, hx_s, hx_uv⟩ := h_inter_nonempty
  rw [huv_empty] at hx_uv
  exact hx_uv
```

Map

Step	Content
A.2.1	Assume no segment point hits the frontier; goal becomes `False`
A.2.2	Covered by $\text{int}\,\Omega \cup \text{int}(\Omega^c)$ (`frontier_eq_for_closed`)
A.2.3	Both opens meet the segment ($\delta \in \text{int}\,\Omega$, $\delta + t_{\text{out}} v \in \text{int}(\Omega^c)$) and are disjoint
A.2.4	`segment_is_connected` (`BasicLemmas.lean:46`) forces a point in both — in $\emptyset$

Step 3: Escape — slice wrappers `Psr` / `Psc` (Lean + map)

PreliminaryLemmas.lean:371-384

```
lemma exists_boundary_point_in_Psr {n : ℕ} (P : Polytope n) (r : ℝ) (δ : CoeffVec n)
    (hδ_in_Ω : δ ∈ P.Ω) (hδ_in_Psr : δ ∈ PsrSet n r)
    (affΩ : AffineSubspace ℝ (CoeffVec n)) (hδ_aff : δ ∈ affΩ)
    (hA_dim : dim (meetDir n (P_sr n r) affΩ) ≥ 1) :
    ∃ δ_bound, δ_bound ∈ PsrSet n r ∩ frontier P.Ω :=
  exists_boundary_point_in_submodule P (P_sr n r) δ hδ_in_Ω hδ_in_Psr affΩ hδ_aff hA_dim
```

A.3

```
/-- Complex case (U = P_sc n s): thin wrapper over the generic lemma. -/
lemma exists_boundary_point_in_Psc {n : ℕ} (P : Polytope n) (s : ℂ) (δ : CoeffVec n)
    (hδ_in_Ω : δ ∈ P.Ω) (hδ_in_Psc : δ ∈ PscSet n s)
    (affΩ : AffineSubspace ℝ (CoeffVec n)) (hδ_aff : δ ∈ affΩ)
    (hA_dim : dim (meetDir n (P_sc n s) affΩ) ≥ 1) :
    ∃ δ_bound, δ_bound ∈ PscSet n s ∩ frontier P.Ω :=
  exists_boundary_point_in_submodule P (P_sc n s) δ hδ_in_Ω hδ_in_Psc affΩ hδ_aff hA_dim
```

A.3

Map

Step	Content
A.1	`ray_escapes_of_bounded` (:73): t_{out} = (|C|+1)/|v| outside S
A.2	`segment_boundary_intersection` (:273): two-opens cover ⇒ $\partial\Omega$ hit (27j–27k)
A.3	`_in_Psr` / `_Psc` (:371,379): U := P_{s_r} / P_{s_c} (this slide)

Step 4: Exposed face — Lean + map

Claim. Let $\delta_{\text{bound}} \in \text{frontier}\, P.\Omega \cap \text{PsrSet}\, n\, r == \{\, \delta \in \mathbb{R}^{n+1} \mid P_\delta(r) = 0 \,\} = \{\, \delta \mid \delta_0 + \delta_1 r + \cdots + \delta_n r^n = 0 \,\}$ with $(\text{interior}\, P.\Omega).\text{Nonempty}$. Then there exists F with $\text{IsExposedFace}\, P\, F \wedge \delta_{\text{bound}} \in F \wedge (r : \mathbb{C}) \in \text{RootSpaceSet}\, F$.

ExposedFaceLemmas.lean:96-136

```
lemma exists_exposed_face_containing_boundary_point {n : ℕ} (P : Polytope n)
    (r : ℝ) (δ_bound : CoeffVec n)
    (hδ_bound_front : δ_bound ∈ frontier P.Ω)
    (hδ_bound_Psr : δ_bound ∈ PsrSet n r)
    (h_int_nonempty : (interior P.Ω).Nonempty) :
    ∃ F : Set (CoeffVec n), IsExposedFace P F ∧ δ_bound ∈ F ∧ (r : ℂ) ∈ RootSpaceSet F := by
  have hδ_bound_in_Ω : δ_bound ∈ P.Ω := frontier_point_in_Ω P δ_bound hδ_bound_front
  have hδ_bound_not_int : δ_bound ∉ interior P.Ω :=
    frontier_point_not_interior P δ_bound hδ_bound_front                          4.1
  have h_convex : Convex ℝ P.Ω := convex_convexHull ℝ _
  have h_int_convex : Convex ℝ (interior P.Ω) := h_convex.interior
  have h_int_open : IsOpen (interior P.Ω) := isOpen_interior                      4.2
  obtain ⟨f, hf_strict⟩ :=
    geometric_hahn_banach_open_point h_int_convex h_int_open hδ_bound_not_int     4.3
  have hf_ne : f ≠ 0 := supporting_func_nonzero P f δ_bound hf_strict h_int_nonempty 4.4
  let c : ℝ := f δ_bound                                                          4.5
  let f_lin : CoeffVec n →ₗ[ℝ] ℝ := f.toLinearMap
  have hf_lin_ne : f_lin ≠ 0 := by
    intro heq
    apply hf_ne
    ext x
    have : f_lin x = 0 := by
      rw [heq]
      simp
    exact this                                                                    4.4
  have hc_upper : ∀ x ∈ P.Ω, f_lin x ≤ c :=
    supporting_hyperplane_upper_bound P f c hf_strict h_int_nonempty              4.6
  let hp : SupportingHyperplane P := {
    f           := f_lin
    c           := c
    nonzero     := hf_lin_ne
    upper_bound := hc_upper
    touches     := ⟨δ_bound, hδ_bound_in_Ω, rfl⟩
  }                                                                               4.7
  have hδ_in_face : δ_bound ∈ ExposedFace hp := by
    unfold ExposedFace
    simp only [Set.mem_setOf_eq]
    exact ⟨hδ_bound_in_Ω, rfl⟩                                                    4.8
  have hr_in_rootspace : (r : ℂ) ∈ RootSpaceSet (ExposedFace hp) :=
    rootspace_mem_of_eval_zero r δ_bound hδ_bound_Psr (ExposedFace hp) hδ_in_face 4.9
  exact ⟨ExposedFace hp, ⟨hp, rfl⟩, hδ_in_face, hr_in_rootspace⟩                  4.10
```

Map

Step	Content
4.1	$\delta_{\text{bound}} \in \Omega \setminus \text{int}\, \Omega$ **(frontier points lie in Ω but never inside)**
4.2	$\text{int}\, \Omega$ **is convex and open: the Hahn–Banach premises**
4.3	**Geometric Hahn–Banach (open convex set vs. outside point) gives continuous linear f with $f < f(\delta_{\text{bound}})$ on the interior**
4.4	***Claim:*** **$f \neq 0$, and its underlying linear map $f_{\text{lin}} : \mathbb{R}^{n+1} \to \mathbb{R}$, $f_{\text{lin}}(x) := f(x)$, likewise satisfies $f_{\text{lin}} \neq 0$. , so if f = 0, then we get 0 < 0 , which is a contradiction**
4.5	**let** $c := f(\delta_{\text{bound}})$
4.6	$f < c$ **on the interior gives** $f \leq c$ **on all of** Ω **(closure plus closed sublevel set)**
4.7	**we construct the hyperplane** $(f, c, \text{nonzero}, \text{upper bound}, \text{touches})$
4.8	$\delta_{\text{bound}} \in \Omega$ **with** $f(\delta_{\text{bound}}) = c$ **puts it in** $F = \Omega \cap \{f = c\}$
	$(r : \mathbb{C}) \in R(F) \stackrel{\text{def}}{=} \{\, s \in \mathbb{C} \mid \exists \delta \in F,\ P_\delta(s) =$

Step 5: Descend — Lean + map (1/2)

Claim. Let $\Omega \subset \mathbb{R}^{n+1}$ be a polytope with $n \geq 1$, and let $F \subseteq \mathbb{R}^{n+1}$ be an exposed face of Ω with $\dim \operatorname{aff}(F) \geq 2$. If $r \in \mathbb{R}$ satisfies $(r : \mathbb{C}) \in R(F)$, then there exists an exposed edge E of Ω with $(r : \mathbb{C}) \in R(E)$.

EdgeDescent.lean:691-715 — part 1, 5.1–5.6 highlighted

```
lemma descend_to_exposed_edge {n : ℕ} (hn : n ≥ 1) (P : Polytope n) (r : ℝ)
    (F : Set (CoeffVec n)) (hF_exp : IsExposedFace P F)
    (hs_F : (r : ℂ) ∈ RootSpaceSet F)
    (hF_dim_ge_2 : Module.finrank ℝ (affineSpan ℝ F).direction ≥ 2) :
    ∃ E, IsExposedEdge P E ∧ (r : ℂ) ∈ RootSpaceSet E := by
  let m_F := Module.finrank ℝ (affineSpan ℝ F).direction
  by_cases hm_F_1 : m_F = 1
  · refine ⟨F, isExposedEdge_of_dim_1 hF_exp hm_F_1, hs_F⟩
  · have hm_F_ge_2 : m_F ≥ 2 := by omega
    obtain ⟨δ_F, hδ_F_in_F, hδ_F_root⟩ :
      ∃ δ ∈ F, ((polyOfVec δ).map (algebraMap ℝ ℂ)).IsRoot (r : ℂ) := hs_F                5.1
    let affF := affineSpan ℝ F
    have hδ_F_Psr : δ_F ∈ PsrSet n r := mem_P_sr_of_isRoot r δ_F hδ_F_root                5.1
    have h_inter_dim : Module.finrank ℝ (meetDir n (P_sr n r) affF) ≥ 1 :=
      intersection_affine_dim_ge_one (P_sr n r) affF δ_F hδ_F_Psr
        (subset_affineSpan ℝ F hδ_F_in_F) (P_sr_dimension r) hm_F_ge_2                   5.2
    obtain ⟨δ_bound, hδ_bound_inter, hδ_bound_front, hδ_bound_not_relint⟩ :=
      exists_boundary_point_in_face_rootspace P r δ_F F hF_exp hδ_F_in_F hδ_F_Psr h_inter_dim   5.3
    have hδ_bound_in_F : δ_bound ∈ F := hδ_bound_inter.1
    have hδ_bound_Psr : δ_bound ∈ (P_sr n r : Set (CoeffVec n)) := hδ_bound_inter.2   5.5
    obtain ⟨G, hG_exp, hδ_bound_in_G, hG_dim_lt⟩ :=
      exists_proper_subface_of_boundary_point P F hF_exp δ_bound
        hδ_bound_in_F hδ_bound_front hδ_bound_not_relint hm_F_ge_2                       5.4
    have hs_G : (r : ℂ) ∈ RootSpaceSet G :=
      rootspace_mem_of_eval_zero r δ_bound hδ_bound_Psr G hδ_bound_in_G                  5.6
```

Map

Step	Content
5.1	$(r : \mathbb{C}) \in R(F)$ **gives** $\delta_F \in F$ **in the zero set** $U = \{\delta \mid P_\delta(r) = 0\}$.
5.2	$\dim U = n$ **and** $\dim \operatorname{aff} F \geq 2$ **give a nonzero root-preserving direction in** $\operatorname{aff} F$ **(Grassmann).**
5.3	**Follow it to** $\delta_{\text{bound}} \in F$ **with** $P_{\delta_{\text{bound}}}(r) = 0$ **on the relative boundary.**
5.4	**A supporting hyperplane there cuts out a proper exposed** $G \subset F$ **(**$\dim G < \dim F$**).**
5.5	$\delta_{\text{bound}} \in G$ **keeps the vanishing.**
5.6	**Hence** $(r : \mathbb{C}) \in R(G)$ **— the root moves to the smaller face.**

Step 5: Descend — Lean + map (2/2)

EdgeDescent.lean:716-733 — part 2, 5.7–5.10 highlighted

```
    by_cases hG_dim_ge_2 : Module.finrank ℝ (affineSpan ℝ G).direction ≥ 2
    · exact descend_to_exposed_edge hn P r G hG_exp hs_G hG_dim_ge_2                5.8
    · by_cases hG_dim_1 : Module.finrank ℝ (affineSpan ℝ G).direction = 1
      · refine (G, isExposedEdge_of_dim_1 hG_exp hG_dim_1, hs_G)                   5.9
      · -- dim(G) = 0: G is the singleton {δ_bound}; get an edge through it.
        have hG_dim_0 : Module.finrank ℝ (affineSpan ℝ G).direction = 0 := by
          have h_not_ge_2 : ¬ Module.finrank ℝ (affineSpan ℝ G).direction ≥ 2 := hG_dim_ge_2
          have h_not_1 : ¬ Module.finrank ℝ (affineSpan ℝ G).direction = 1 := hG_dim_1
          omega
        have hG_single : G = {δ_bound} :=
          exposed_face_dim_zero_singleton G hG_dim_0 δ_bound hδ_bound_in_G
        obtain ⟨E, hE_edge, hδ_in_E⟩ :=
          exists_exposed_edge_of_singleton hn P δ_bound G hG_exp hG_single
        have h_root_E : (r : ℂ) ∈ RootSpaceSet E :=
          rootspace_mem_of_eval_zero r δ_bound hδ_bound_Psr E hδ_in_E
        exact ⟨E, hE_edge, h_root_E⟩                                              5.10
  termination_by Module.finrank ℝ (affineSpan ℝ F).direction                      5.7
  decreasing_by exact hG_dim_lt
```

Map

Step	Content
5.7	Each descent step replaces F by G with $\dim G < \dim F$. Hence the dimension drops by at least one at every iteration. Therefore the process can continue only finitely many times.
5.8	If $\dim G \geq 2$, then G satisfies the same hypothesis as F. Hence the same argument applies with F replaced by G. This is the recursive descent step.
5.9	If $\dim G = 1$, then G is a segment, i.e. an exposed edge. Since $(r : \mathbb{C}) \in R(G)$ already holds, the descent stops here. Take $E = G$ to conclude.
5.10	If $\dim G = 0$, then G is the singleton $\{\delta_{\text{bound}}\}$. Choose an exposed edge E through this vertex, so $\delta_{\text{bound}} \in E$. Since $P_{\delta_{\text{bound}}}(r) = 0$, the root transfers to E and the proof is complete.

Step 6: Degenerate cases — Lean + map

Claim. Let $m := \dim \operatorname{aff}(\Omega)$. Then $m = 0$ is impossible ($\operatorname{int} \Omega \neq \emptyset$), and $m = 1$ makes Ω itself the edge.

Proof. **D.1** $m = 0$: `polytope_direction_dim_pos` gives $\dim \geq 1$, contradiction. **D.2** $m = 1$: `polytope_dim1_is_exposed_edge` makes Ω itself the edge; $s = \bar{r}$ moves the root witness to E.

Lemma61.lean:191-203 — degenerate cases, ①② highlighted

```
· by_cases hm0 : m = 0                                                    ①
  · have h_pos : Module.finrank ℝ (affineSpan ℝ P.Ω).direction ≥ 1 :=
      polytope_direction_dim_pos P
    omega
  · have hm1 : m = 1 := by omega                                          ②
    have h_Ω_is_edge : IsExposedEdge P P.Ω :=
      polytope_dim1_is_exposed_edge hn P hm1
    refine ⟨P.Ω, h_Ω_is_edge, ?_⟩
    rw [hs_real]
    exact rootspace_mem_of_eval_zero s.re δ
      (mem_P_sr_of_isRoot s.re δ (by
        rw [← hs_real]
        exact hδ_root)) P.Ω hδ_in_Ω
```

Map

Step	Content
D.1 ①	$m = 0$ **contradicts nonempty interior (**$\dim \geq 1$**)**
D.2 ②	$m = 1$**: Ω itself the edge;** $s = \bar{r}$ **moves the root** $\Omega \to E$

Lemma 6.1 (real case) — recap

1. **Step 1** unpacked $s \in R(\Omega)$ to $\delta \in P_{s_r}(r)$ with $s = \bar{r}$.
2. **Step 2** gave $\dim(U \cap \operatorname{aff}\Omega) \geq 1$.
3. **Step 3** followed it to $\delta_{\text{bound}} \in P_{s_r} \cap \partial\Omega$.
4. **Step 4** separated at δ_{bound}: exposed $F \ni \delta_{\text{bound}}$ with $\bar{r} \in R(F)$.
5. **Step 5** descended $F \to G \to \cdots \to E$ with $\bar{r} \in R(E)$, $\dim E = 1$.

Degenerate cases (`lemma61_real:191-203`): $m = 0$ contradicts nonempty interior; $m = 1$ makes Ω itself the edge (`polytope_dim1_is_exposed_edge`).

Lemma 6.1: complex case — what is extra

Mathematics

- **Claim.** If $s \in R(\Omega)$ with $\mathrm{Im}(s) \neq 0$, then $s \in R(F)$ for an exposed face F with $\dim \mathrm{aff}(F) \leq 2$ — a face, not an edge.
- **Thinner slice.** $U = \{\delta \mid P_\delta(s) = 0\}$ has dimension $n - 1$: real and imaginary parts must each vanish — two equations cut twice.
- **Stronger count.** Grassmann needs direction dimension ≥ 3 (and $n \geq 2$).
- **Weaker target.** One condition gives a line to walk to an edge; two give isolated points that cannot slide — descent stops at dimension 2.
- **New degenerate case $n = 1$.** Linear P_δ forces $s = -\delta_0/\delta_1 \in \mathbb{R}$ unless $\delta_1 = 0$, which Assumption 6.1 forbids.

Lean

- `P_sc_dimension` (`hn : n ≥ 1`, `hs : s.im ≠ 0`): the complex slice has dimension $n - 1$ (evaluation onto $\mathbb{C}$ is surjective of real rank 2).
- `mem_P_sc_of_isRoot`, `intersection_affine_dim_ge_one_complex` (needs direction dimension ≥ 3), `exists_boundary_point_in_Psc`: Steps 1–3 with P_{s_c} in place of P_{s_r}.
- `exists_exposed_face_containing_boundary_point_complex`: Step 4 for the complex slice.
- `descend_to_exposed_face_with_dim`: Step 5 stopping at dimension ≤ 2.
- Assumption 6.1 as explicit hypothesis `hleading : ∀ v ∈ P.vertices, 0 < v (Fin.last n)`, threaded through `lemma61_complex_with_dim` → `lemma61_complex` → `lemma61`; the $n = 1$ case closes by leading-coefficient positivity (`convexHull_min`).

Section 5 · Lemma 6.2, real case

Lemma 6.2 (real case) — overview

> **Theorem (Lemma 6.2).** Let F be an exposed face of Ω with $\dim \operatorname{aff}(F) = 2$. Then $\partial R(F) \subseteq R(\partial F)$: every boundary point of the root space of F is a root on the relative boundary (`lemma62`, `Lemma62.lean:1446`).

The Lean proof splits on `s_star.im = 0` (`by_cases hreal`): the **real case** (`lemma62_real_case`, this section) goes through P_{s_r} as in Lemma 6.1; the **complex case** (`lemma62_complex_case`) needs the W-matrix machinery — future work.

Lemma 6.2 (real case) — goal

> **Goal.** Let F be exposed with $\dim \operatorname{aff}(F) = 2$ and $s^\star \in \operatorname{frontier} R(F)$ real ($\operatorname{Im}(s^\star) = 0$). Then $s^\star \in R(\operatorname{relativeBoundary} F)$ (`lemma62_real_case`, `Lemma62.lean:127`).

Proof sketch. **6.1–6.3** $R(F)$ is closed, so $s^\star \in R(F)$ with $s^\star = \bar{r}$. **6.4–6.5** Unpack $\delta^\star \in F$; split on relint. **6.6–6.8** Escape to $\delta_{\text{bound}} \in F \cap P_{s_r}$ off the relint. **6.9–6.12** Transfer the root; close Case A the same way.

`PreliminaryLemmas.lean:422-423`

```
def relativeBoundary (F : Set (CoeffVec n)) : Set (CoeffVec n) :=
  F \ intrinsicInterior ℝ F
```

Lemma 6.2 (real case) — math proof (1/2)

6.1 F is compact and convex

- `isExposedFace_isCompact`, `isExposedFace_convex`: an exposed face inherits both from $P.\Omega$.

6.2 $s^\star$ is real

- `Complex.ext` with `hreal`: $s^\star = \bar{r}$, $r = \mathrm{Re}(s^\star)$.

6.3 $R(F)$ is closed, so $s^\star \in R(F)$

- `rootSpaceSet_isClosed_of_isCompact`: zero set of a jointly continuous evaluation, projected from compact F.
- $\mathrm{frontier}\, R(F) \subseteq \overline{R(F)} = R(F)$ (`frontier_subset_closure`, `IsClosed.closure_eq`).

6.4 Unpack to $\delta^\star$

- $s^\star \in R(F)$ gives $\delta^\star \in F$ with root; `mem_P_sr_of_isRoot` (+ rewrite by 6.2) puts $\delta^\star \in P_{s_r}(r)$.

6.5 Split on the relative interior

- `by_cases hδ_star_relint`: $\delta^\star \notin \mathrm{intrinsicInterior}\, F$ (Case A, immediate) vs $\in$ (Case B, escape below).

6.6 Dimension count gives v

- $U = P_{s_r}$ ($\dim n$), $V = \mathrm{dir}$ ($\dim \geq 2$ from hF_dim_2): `finrank_inf_ge_one` gives $\dim(U \sqcap V) \geq 1$, hence nonzero v (`exists_ne`).

Lemma 6.2 (real case) — math proof (2/2)

6.7 Escape and the set S

- Split v into U- and V-components; `ray_escapes_of_bounded` gives t_{out} with $\delta^\star + t_{out} v \notin F$ (bounded Ω forces an exit point).
- $S = \{t \mid 0 \le t \wedge \delta^\star + tv \in F\}$: nonempty ($0$), closed (preimage of closed F under a continuous ray $\cap$ `Ici`), bounded above by t_{out} via convexity (helper `ray_stay_set_bddAbove`).

6.8–6.10 Supremum gives δ_{bound}

- $t_1 = \operatorname{sSup} S \in S$ (`IsClosed.csSup_mem`): $\delta_{\text{bound}} = \delta^\star + t_1 v \in F \cap P_{s_r}$ (`add_mem`, `smul_mem`).
- `not_mem_intrinsicInterior_of_escapes_along_direction`: $\delta_{\text{bound}} \notin \operatorname{intrinsicInterior} F$.

6.11 The root transfers

- $\delta_{\text{bound}} \in \operatorname{relativeBoundary} F = F \setminus \operatorname{intrinsicInterior}$; `rootspace_mem_of_eval_zero` gives $\bar{r} \in R(\operatorname{relativeBoundary} F)$; `rw [h_s_star_eq]` turns $\bar{r}$ back into $s^\star$.

6.12 Case A closes the same way

- $\delta^\star$ itself is already on the relative boundary; same transfer, same rewrite. ∎

Lemma 6.2 (real case) — helper `ray_stay_set_bddAbove` (Lean)

Lemma62.lean:91-121 — helper lemma, no highlights

```
lemma ray_stay_set_bddAbove {n : ℕ} {F : Set (CoeffVec n)} {δ_star v : CoeffVec n}
    (hF_convex : Convex ℝ F) (hδ_star_in_F : δ_star ∈ F)
    (t_out : ℝ) (ht_out_pos : 0 < t_out) (ht_out : δ_star + t_out • v ∉ F) :
    BddAbove {t : ℝ | 0 ≤ t ∧ δ_star + t • v ∈ F} := by
  refine ⟨t_out, ?_⟩
  rintro t ⟨ht_nonneg, ht_mem⟩
  by_contra! h_gt
  have ha_nonneg : 0 ≤ t_out / t := div_nonneg (by linarith) (by linarith)
  have hdiv : t_out / t ≤ 1 := (div_le_one (by linarith)).mpr (by linarith)
  have hb_nonneg : 0 ≤ 1 - t_out / t := by linarith
  have hsum : (t_out / t : ℝ) + (1 - t_out / t) = 1 := by ring
  have hstar : StarConvex ℝ (δ_star + t • v) F := hF_convex ht_mem
  have h_conv : ((t_out / t : ℝ) • (δ_star + t • v) + (1 - t_out / t) • δ_star) = δ_star + t_out • v := by
    calc
      (t_out / t : ℝ) • (δ_star + t • v) + (1 - t_out / t) • δ_star
          = (t_out / t) • δ_star + (t_out / t) • (t • v) + (1 - t_out / t) • δ_star := by rw [smul_add]
      _ = ((t_out / t) • δ_star + (1 - t_out / t) • δ_star) + (t_out / t) • (t • v) := by abel
      _ = ((t_out / t + (1 - t_out / t)) • δ_star) + ((t_out / t) * t) • v := by
        simp [smul_smul]
      _ = (1 • δ_star) + (t_out • v) := by
        have h_t_ne_zero : t ≠ 0 := by linarith
        have h_sum : t_out / t + (1 - t_out / t) = 1 := by ring
        have h_mul : (t_out / t) * t = t_out := by field_simp [h_t_ne_zero]
        simp [h_sum, h_mul]
      _ = δ_star + t_out • v := by simp
  have h_mem_conv : (t_out / t : ℝ) • (δ_star + t • v) + (1 - t_out / t) • δ_star ∈ F :=
    hstar hδ_star_in_F ha_nonneg hb_nonneg hsum
  have h_mem : δ_star + t_out • v ∈ F := by
    rw [← h_conv]
    exact h_mem_conv
  exact ht_out h_mem
```

Lemma 6.2 (real case) — walkthrough 6.1–6.6

Lemma62.lean:127-162, 6.1–6.6 highlighted

```
theorem lemma62_real_case {n : ℕ}  (P : Polytope n)
    (F : Set (CoeffVec n)) (hF_exp : IsExposedFace P F)
    (hF_dim_2 : dim (affineSpan ℝ F).direction = 2)
    (s_star : ℂ) (hs_star_front : s_star ∈ frontier (RootSpaceSet F)) (hreal : s_star.im = 0) :
    s_star ∈ RootSpaceSet (relativeBoundary F) := by
  have hF_compact : IsCompact F := isExposedFace_isCompact P hF_exp
  have hF_convex : Convex ℝ F := isExposedFace_convex P hF_exp                    6.1
  have h_s_star_eq : s_star = (s_star.re : ℂ) := by
    apply Complex.ext <;> simp [hreal]                                           6.2
  have h_root_closed : IsClosed (RootSpaceSet F) :=
    rootSpaceSet_isClosed_of_isCompact hF_compact
  have hs_star_in_RF : s_star ∈ RootSpaceSet F := by
    have h_sub : frontier (RootSpaceSet F) ⊆ RootSpaceSet F := by
      calc
        frontier (RootSpaceSet F) ⊆ closure (RootSpaceSet F) := frontier_subset_closure
        _ = RootSpaceSet F := h_root_closed.closure_eq
    exact h_sub hs_star_front                                                    6.3
  rcases hs_star_in_RF with ⟨δ_star, hδ_star_in_F, hδ_star_root⟩
  have hδ_star_in_Psr : δ_star ∈ PsrSet n s_star.re :=
    mem_P_sr_of_isRoot s_star.re δ_star (by
      rw [h_s_star_eq] at hδ_star_root
      exact hδ_star_root)                                                        6.4
  by_cases hδ_star_relint : δ_star ∈ intrinsicInterior ℝ F
  · let U : Submodule ℝ (CoeffVec n) := P_sr n s_star.re
    let V : Submodule ℝ (CoeffVec n) := (affineSpan ℝ F).direction               6.5
    have hU_dim : dim U = n := P_sr_dimension s_star.re
    have hV_dim_ge_2 : dim V ≥ 2 := by rw [hF_dim_2]
    have h_inter_dim_ge_1 : dim (U ⊓ V) ≥ 1 :=
      finrank_inf_ge_one U V hU_dim hV_dim_ge_2
    have h_inter_finrank_pos : 0 < dim (U ⊓ V) := by omega
    have h_inter_nontrivial : Nontrivial ↥(U ⊓ V) :=
      Module.nontrivial_of_finrank_pos h_inter_finrank_pos
    obtain ⟨v_sub, hv_sub_ne⟩ := exists_ne (0 : ↥(U ⊓ V))
    let v : CoeffVec n := v_sub.val
    have hv_ne : v ≠ 0 := by
      intro h; apply hv_sub_ne; exact Subtype.ext h                              6.6
```

Map

Step	Content
6.1	F **compact and convex from** $P.\Omega$
6.2	$\mathrm{Im}(s^\star) = 0 \implies s^\star = \bar{r}$
6.3	$R(F)$ **closed, so** $s^\star \in \mathrm{frontier}\, R(F) \subseteq R(F)$
6.4	$\delta^\star \in F$ **with root** $\implies \delta^\star \in P_{s_r}(r)$
6.5	**Split: off-relint (Case A) vs on-relint (Case B)**
6.6	$U = P_{s_r}$ **(**$\dim n$**),** $V = \mathrm{dir}$ **(**$\dim \geq 2$**) give nonzero** v

Lemma 6.2 (real case) — walkthrough 6.7–6.9

Lemma62.lean:163-190, 6.7–6.9 highlighted

```
    have hv_U : v ∈ U := by
      have h := v_sub.property
      rw [Submodule.mem_inf] at h
      exact h.1
    have hv_V : v ∈ V := by
      have h := v_sub.property
      rw [Submodule.mem_inf] at h
      exact h.2
    have hv_affF_dir : v ∈ (affineSpan ℝ F).direction := hv_V
    obtain ⟨t_out, ht_out_pos, ht_out⟩ :=
      ray_escapes_of_bounded hF_compact.isBounded δ_star v hδ_star_in_F hv_ne
    let S : Set ℝ := {t | 0 ≤ t ∧ δ_star + t • v ∈ F}
    have hS_nonempty : S.Nonempty := ⟨0, ⟨by norm_num, by simpa using hδ_star_in_F⟩⟩
```

6.7

```
    have hS_closed : IsClosed S := by
      have h_cont : Continuous (fun (t : ℝ) => δ_star + t • v) := by
        refine Continuous.add continuous_const ?_
        exact Continuous.smul continuous_id continuous_const
      have h_preimage_closed : IsClosed {t | δ_star + t • v ∈ F} :=
        hF_compact.isClosed.preimage h_cont
      have h_nonneg_closed : IsClosed {t : ℝ | 0 ≤ t} := isClosed_Ici
      have hS_eq : S = {t | δ_star + t • v ∈ F} ∩ {t : ℝ | 0 ≤ t} := by
        ext t; constructor
        · rintro ⟨ht_nonneg, ht_mem⟩; exact ⟨ht_mem, ht_nonneg⟩
        · rintro ⟨ht_mem, ht_nonneg⟩; exact ⟨ht_nonneg, ht_mem⟩
      rw [hS_eq]
      exact h_preimage_closed.inter h_nonneg_closed
```

6.8

```
    have h_bdd_above : BddAbove S :=
      ray_stay_set_bddAbove hF_convex hδ_star_in_F t_out ht_out_pos ht_out
```

6.9

Map

Step	Content
6.7	v splits into slice and direction parts; t_{out} escapes F; ray-stay set nonempty at $t = 0$
6.8	S closed: continuous ray meets closed F
6.9	S bounded above by t_{out} via convexity (helper `ray_stay_set_bddAbove`)

Lemma 6.2 (real case) — walkthrough 6.10–6.12

Lemma62.lean:191-220, 6.10–6.12 highlighted

```
    let t1 := sSup S
    have h_max : t1 ∈ S := by
      simpa [t1] using hS_closed.csSup_mem hS_nonempty h_bdd_above
    rcases h_max with ⟨h_t1_nonneg, h_t1_mem⟩
    let δ_bound : CoeffVec n := δ_star + t1 • v
    have hδ_bound_in_F : δ_bound ∈ F := h_t1_mem
    have hv_in_Psr : v ∈ (P_sr n s_star.re : Set (CoeffVec n)) := hv_U
    have hδ_bound_in_Psr : δ_bound ∈ (P_sr n s_star.re : Set (CoeffVec n)) := by
      dsimp [δ_bound]
      apply Submodule.add_mem (P_sr n s_star.re)
      · exact hδ_star_in_Psr
      · exact Submodule.smul_mem (P_sr n s_star.re) t1 hv_in_Psr
```

6.10

```
    have h_not_relint : δ_bound ∉ intrinsicInterior ℝ F :=
      not_mem_intrinsicInterior_of_escapes_along_direction
        F hF_convex δ_star hδ_star_in_F v hv_ne hv_affF_dir
        S rfl hS_nonempty h_bdd_above
        t1 rfl h_t1_nonneg h_t1_mem
        t_out ht_out_pos ht_out
    have hδ_bound_rel_boundary : δ_bound ∈ relativeBoundary F :=
      ⟨hδ_bound_in_F, h_not_relint⟩
    have h_r_in_RF : (s_star.re : ℂ) ∈ RootSpaceSet (relativeBoundary F) :=
      rootspace_mem_of_eval_zero s_star.re δ_bound hδ_bound_in_Psr (relativeBoundary F) hδ_bound_rel_boundary
    rw [h_s_star_eq]
    exact h_r_in_RF
```

6.1

```
  · have hδ_star_rel_boundary : δ_star ∈ relativeBoundary F :=
      ⟨hδ_star_in_F, hδ_star_relint⟩
    have h_r_in_RF : (s_star.re : ℂ) ∈ RootSpaceSet (relativeBoundary F) :=
      rootspace_mem_of_eval_zero s_star.re δ_star hδ_star_in_Psr (relativeBoundary F) hδ_star_rel_boundary
    rw [h_s_star_eq]
    exact h_r_in_RF
```

6.12

Map

Step	Content
6.10	Supremum stays in closed S; $\delta_{\text{bound}} \in F \cap P_{s_r}$
6.11	δ_{bound} off relint; root transfers to the relative boundary
6.12	$\delta^\star$ itself works; same transfer

Lemma 6.2 (real case) — proof map (1/2)

Step	Content
6.1	F compact and convex
6.2	$s^\star = \bar{r}$
6.3	$R(F)$ closed $\implies$ $s^\star \in R(F)$
6.4	$\delta^\star \in F \cap P_{s_r}(r)$
6.5	Case A (off relint) vs Case B (escape)
6.6	$U \sqcap V$ nontrivial $\to$ nonzero v
helpers	Step-2 formula + ray escape

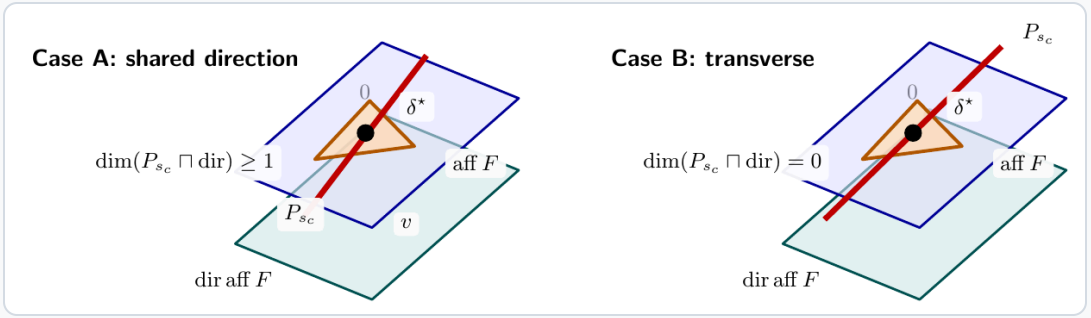
Lemma 6.2 (real case) — proof map (2/2)

Step	Content
6.7	Direction split, escape time, ray-stay set
6.8	S closed (preimage $\cap$ `Ici`)
6.9	S bounded above by t_{out} (helper `ray_stay_set_bddAbove`)
6.10	Supremum gives $\delta_{\text{bound}} \in F \cap P_{s_r}$
6.11	Off relint; root moves, rewrite to $s^\star$
6.12	$\delta^\star$ itself; same transfer
dispatcher	`lemma62` (`Lemma62.lean:1446`)
complex twin	`lemma62_complex_case` (`Lemma62.lean:1164`)

Lemma 6.2: complex case — what is extra

Mathematics

- **Claim.** Same goal for non-real $s^\star$: a boundary root of $R(F)$ already lives on the relative boundary of F (complex slice P_{s_c} in place of P_{s_r}).
- **New split.** The slice $P_{s_c}(s^\star)$ and the face directions $\operatorname{dir}\operatorname{aff} F$ generically meet only in $\{0\}$ — no direction to walk. Case A shares a nonzero direction v; Case B ($\dim = 0$) is transverse.
- **Case A ($\dim \geq 1$).** Walk v to the relative boundary and transfer the root — the real-case escape.
- **Case B ($\dim = 0$).** Non-roots $s_k \to s^\star$; Cramer on V_0, V_1 ($\det \neq 0$) gives $\delta_k \to \delta^\star$ in $\operatorname{aff}(F)$ carrying s_k.
- **Limit.** $\delta_k \notin F$ yet $\delta_k \to \delta^\star$ rules out interior — $\delta^\star$ is on the relative boundary carrying $s^\star$.



Lean

- Same opening via `isExposedFace_isCompact`, `rootSpaceSet_isClosed_of_isCompact`, `mem_P_sc_of_isRoot`.
- Case A: `exists_boundary_point_in_face_rootspace_complex` (via `h_inter_dim_of_meet`), then `rootspace_mem_of_isRoot` — the real-case escape replayed in P_{s_c}.
- Case B: `exists_seq_notin_RootSpaceSet_tendsto` (the s_k); `faceCols` basis with `detOf_ne_zero_of_transverse` + `tendsto_detOf` (determinant stays nonzero); Cramer corrections `lam0` / `lam1` with `hev_root` (δ_k carries s_k while $\det_k \neq 0$); `hdeltak_tendsto` ($\lambda_k \to 0$); the interior assumption contradicts `hev_notF` — hence `hnot_ri`.
- Machinery: quadratic-factor perturbations `Wcol` / `Wmul` staying inside the slice (`evalAtComplex_Wcol`, `Wmul_mem_Psc`, `tendsto_Wmul`); `evalAtComplex` with `tendsto_evalAtComplex`; transverse 2×2 `detOf`.

Edge Theorem — formal proof

> **Theorem (Edge Theorem).** Let $\Omega \subset \mathbb{R}^{n+1}$ ($n \geq 1$) be a polytope satisfying Assumption 6.1. Then every $s \in \partial R(\Omega)$ lies in $R(E)$ for some exposed edge E of Ω.

Proof

1. Let $s \in \partial R(\Omega)$. Since $R(\Omega)$ is closed, $s \in R(\Omega)$ and $s \notin \operatorname{int} R(\Omega)$.
2. If $\operatorname{Im}(s) = 0$, Lemma 6.1 (real case) gives an exposed edge E with $s \in R(E)$. Done.
3. Else $\operatorname{Im}(s) \neq 0$: Lemma 6.1 (complex case) gives an exposed face F with $s \in R(F)$ and $\dim \operatorname{aff}(F) \leq 2$.
4. If $\dim = 2$: $R(F)$ is closed, and $\operatorname{int} R(F) \subseteq \operatorname{int} R(\Omega) \not\ni s$, so $s \in \partial R(F)$. Lemma 6.2 moves s to $R(\operatorname{relativeBoundary} F)$; the witness lies on the relative boundary, giving a proper exposed subface $G \subsetneq F$ ($\dim G \leq 1$) with $s \in R(G)$.
5. If $\dim \leq 1$: the face (G, resp. F) is already an edge ($\dim = 1$) or a singleton ($\dim = 0$), and a singleton face sits on an exposed edge through it. Either way $s \in R(E)$ for an exposed edge E. ∎

Lean (`EdgeTheorem.lean`)

- Closedness + unpacking: `rootSpaceSet_isClosed_of_isCompact`, `frontier_subset_closure`.
- Step 2 is `edge_real` (exactly `lemma61_real`).
- Step 3 is `lemma61_complex_with_dim` (face of dimension at most two).
- Step 4 uses `interior_mono` with $R(F) \subseteq R(\Omega)$ to land in $\partial R(F)$; Lemma 6.2 enters as `face_boundary_subset`.
- The last descent is `exists_proper_subface_of_boundary_point`, closing via `isExposedEdge_of_dim_1` or `exposed_face_dim_zero_singleton` + `exists_exposed_edge_of_singleton` (factored as `exposed_face_root_dim_le1_to_edge`).
- Assembly: `edge_theorem` (`by_cases hreal`, then `by_cases hdim2`).

Edge Theorem — lemmas behind steps 1–5

Step 1. Root spaces over compact sets are closed, so a boundary root is a genuine root outside the interior.

Lemma62.lean:37-38

```
lemma rootSpaceSet_isClosed_of_isCompact {n : ℕ} {W : Set (CoeffVec n)} (hW : IsCompact W) :
    IsClosed (RootSpaceSet W)
```

How: Let $Z = \{(\delta, s) \in W \times \mathbb{C} : \tilde{P}_\delta(s) = 0\}$, the set of polynomial–root pairs. It is closed, being the zero set of the continuous evaluation map. $\pi(\delta, s) = s$ satisfies $\pi(Z) = R(W)$: indeed $s \in \pi(Z)$ iff some $\delta \in W$ vanishes at s. Since W is compact, $\pi : W \times \mathbb{C} \to \mathbb{C}$ is a closed map, so $\pi(Z) = R(W)$ is closed.

Step 2. Every real root of the family already lives on an exposed edge.

EdgeTheorem.lean:37-40

```
theorem edge_real {n : ℕ} (hn : n ≥ 1) (P : Polytope n) (s : ℂ)
    (hs : s ∈ RootSpace P) (hreal : s.im = 0) :
    ∃ E, IsExposedEdge P E ∧ s ∈ RootSpaceSet E
```

How: Lemma 6.1, real case: $s = \bar{r} \in R(\Omega)$ yields E with $s \in R(E)$.

Step 3. Every non-real root lives on an exposed face of dimension at most two.

Lemma61.lean:208-211

```
theorem lemma61_complex_with_dim (hn : n ≥ 1) (P : Polytope n) (s : ℂ) (hs : s ∈ RootSpace P)
    (hleading : ∀ v ∈ P.vertices, 0 < v (Fin.last n)) : s.im ≠ 0 →
    ∃ F, IsExposedFace P F ∧ s ∈ RootSpaceSet F ∧
      Module.finrank ℝ (affineSpan ℝ F).direction ≤ 2
```

How: Steps 1–5 in the $(n-1)$-dimensional slice, descending to $\dim \le 2$ (the count requires $\dim \ge 3$); $n = 1$ excluded by the leading coefficient.

Step 4. Boundary roots of a two-dimensional exposed face live on its relative boundary.

EdgeTheorem.lean:56-60

```
theorem face_boundary_subset {n : ℕ} (hn : n ≥ 1) (P : Polytope n)
    (F : Set (CoeffVec n)) (hF_exp : IsExposedFace P F)
    (hF_dim_2 : dim (affineSpan ℝ F).direction = 2) :
    frontier (RootSpaceSet F) ⊆ RootSpaceSet (relativeBoundary F)
```

How: split on $\mathrm{Im}(s^\star) = 0$ into the real and complex cases.

Step 5. Roots on an at-most-one-dimensional exposed face move to an exposed edge — directly at dimension one, through a singleton vertex at dimension zero.

EdgeTheorem.lean:66-70

```
private theorem exposed_face_root_dim_le1_to_edge {n : ℕ} (hn : n ≥ 1) (P : Polytope n)
    (s : ℂ) (F : Set (CoeffVec n)) (hF_exp : IsExposedFace P F)
    (hsF : s ∈ RootSpaceSet F)
    (hF_dim : Module.finrank ℝ (affineSpan ℝ F).direction ≤ 1) :
    ∃ E, IsExposedEdge P E ∧ s ∈ RootSpaceSet E
```

How: $\dim = 1$ is already an edge; $\dim = 0$ is a singleton lying on an edge through it.

Edge Theorem — Lean implementation

EdgeTheorem.lean:82-89 — statement

```
theorem edge_theorem {n : ℕ} (hn : n ≥ 1) (P : Polytope n) (s : ℂ)
    (hleading : ∀ v ∈ P.vertices, 0 < v (Fin.last n))
    (hs : s ∈ frontier (RootSpace P)) :
    ∃ E, IsExposedEdge P E ∧ s ∈ RootSpaceSet E := by
```

EdgeTheorem.lean:90-121 — proof (1/2)

```
  have hroot_closed : IsClosed (RootSpace P) := by
    change IsClosed (RootSpaceSet P.Ω)
    exact rootSpaceSet_isClosed_of_isCompact P.isCompact
  have hs_mem : s ∈ RootSpace P := by
    have hcl : s ∈ closure (RootSpace P) := frontier_subset_closure hs
    rw [hroot_closed.closure_eq] at hcl
    exact hcl
  have hs_not_int : s ∉ interior (RootSpace P) := by
    change s ∈ closure (RootSpace P) \ interior (RootSpace P) at hs
    exact hs.2
  by_cases hreal : s.im = 0
  · exact edge_real hn P s hs_mem hreal
  · obtain ⟨F, hF_exp, hsF, hF_dim⟩ :=
      lemma61_complex_with_dim hn P s hs_mem hleading hreal
    by_cases hdim2 : Module.finrank ℝ (affineSpan ℝ F).direction = 2
    · have hroot_closed_F : IsClosed (RootSpaceSet F) :=
        rootSpaceSet_isClosed_of_isCompact (isExposedFace_isCompact P hF_exp)
      have hsF_not_int : s ∉ interior (RootSpaceSet F) := by
        intro hInt
        have hsub : RootSpaceSet F ⊆ RootSpace P := by
          intro z hz
          obtain ⟨δ, hδF, hδroot⟩ := hz
          exact rootspace_mem_of_isRoot z δ hδroot P.Ω
            (isExposedFace_subset_Ω hF_exp hδF)
        exact hs_not_int (interior_mono hsub hInt)
      have hsF_front : s ∈ frontier (RootSpaceSet F) := by
        have hsF_mem : s ∈ closure (RootSpaceSet F) := by
          rw [hroot_closed_F.closure_eq]
          exact hsF
        change s ∈ closure (RootSpaceSet F) \ interior (RootSpaceSet F)
        exact ⟨hsF_mem, hsF_not_int⟩
```

EdgeTheorem.lean:122-151 — proof (2/2)

```
      have hs_rel : s ∈ RootSpaceSet (relativeBoundary F) :=
        face_boundary_subset hn P F hF_exp hdim2 hsF_front
      obtain ⟨δ, hδrel, hδroot⟩ := hs_rel
      have hδF : δ ∈ F := hδrel.1
      have hδ_not_relint : δ ∉ intrinsicInterior ℝ F := hδrel.2
      have hF_compact : IsCompact F := isExposedFace_isCompact P hF_exp
      have hδ_not_int : δ ∉ interior F := by
        intro hInt
        exact hδ_not_relint
          (interior_subset_intrinsicInterior (𝕜 := ℝ) (s := F) hInt)
      have hδ_front : δ ∈ frontier F := by
        rw [frontier_eq_for_closed F hF_compact.isClosed]
        exact ⟨hδF, hδ_not_int⟩
      obtain ⟨G, hG_exp, hδG, hG_dim_lt⟩ :=
        exists_proper_subface_of_boundary_point P F hF_exp δ hδF hδ_front
          hδ_not_relint (by omega)
      have hsG : s ∈ RootSpaceSet G :=
        rootspace_mem_of_isRoot s δ hδroot G hδG
      by_cases hG1 : Module.finrank ℝ (affineSpan ℝ G).direction = 1
      · exact ⟨G, isExposedEdge_of_dim_1 hG_exp hG1, hsG⟩
      · have hG0 : Module.finrank ℝ (affineSpan ℝ G).direction = 0 := by omega
        obtain ⟨δ', hδ'G, hδ'root⟩ := hsG
        have hG_single : G = {δ'} :=
          exposed_face_dim_zero_singleton G hG0 δ' hδ'G
        obtain ⟨E, hE, hδ'E⟩ :=
          exists_exposed_edge_of_singleton hn P δ' G hG_exp hG_single
        exact ⟨E, hE, rootspace_mem_of_isRoot s δ' hδ'root E hδ'E⟩
    · -- In dimension zero or one, the exposed face itself is a singleton or edge.
      have hdim_le1 : Module.finrank ℝ (affineSpan ℝ F).direction ≤ 1 := by omega
      exact exposed_face_root_dim_le1_to_edge hn P s F hF_exp hsF hdim_le1
```

Future work — outlook

1. **Lemma 6.2, complex case** (non-real s, `lemma62_complex_case`): only the real case is proved here (Section 5).
2. **Lemma 6.1, complex case** (non-real s, `lemma61_complex`): only the real case is proved here (Section 4).
3. Assembling the full Edge Theorem from the lemmas in Lean (`EdgeTheorem.lean`).